



ELECTRON

BEAM

SYSTEMS



**«Installation for the production of granules from nickel/titanium alloys and intermetallic compound
by centrifugal spraying»**

<EBS-RM>

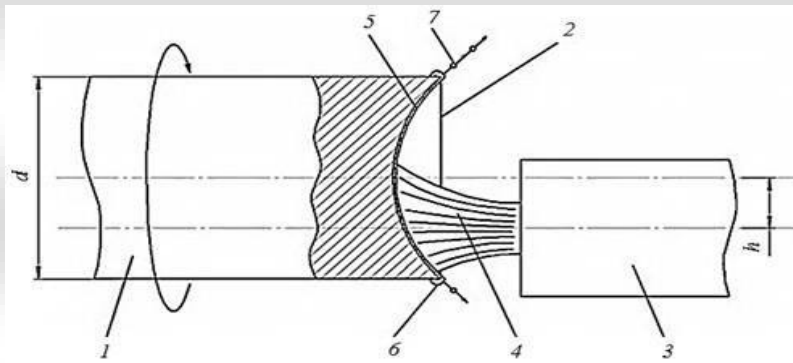


Electronic Beam Systems LLC presents a modern industrial plant for the production of granules from nickel/titanium alloys and intermetallic compounds by centrifugal plasma spraying of a rotating electrode in an inert gas.

Main technical characteristics of the installation:

Parameters:	Unit of measurmen	Data
1	2	3
Productivity (technological):	kg per hour	Up to 25
Material	nickel/titanium alloys and intermetallic compounds	
Dimensions of the remelted workpiece:		
- diameter	m	Φ52-80
- length	m	650-750
- weight	kg	до 32
The rotation frequency of the workpiece (max):	min ⁻¹	38000
- gas	Mixture of gases Ar + He in the ratio ~ 1/9	
Heating module:		
- generator power (max)	kW	200
- generator arc current	A	800...1800
- rated voltage	V	60..120
- cooling water consumption	M³/h	Up to 1,0
Electrical equipment (power supply parametres):		
- voltage	V	380
- current frequency	Hz	50
- installed capacity	kW	Up to 300

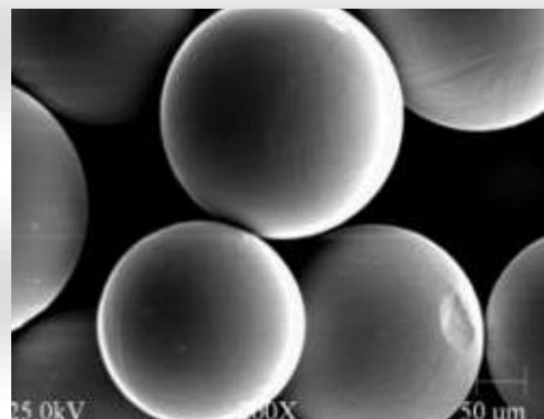
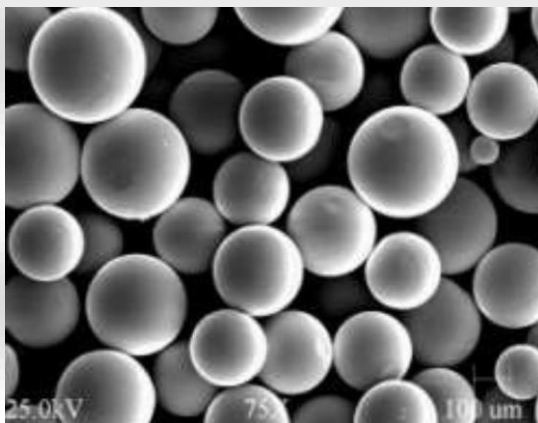
The scheme of production of granules by PREP:



1. cast cylindrical billet
2. the end of the workpiece
3. plasmatron
4. plasma torch
5. melt film
6. the crown of the melt
7. melt drops

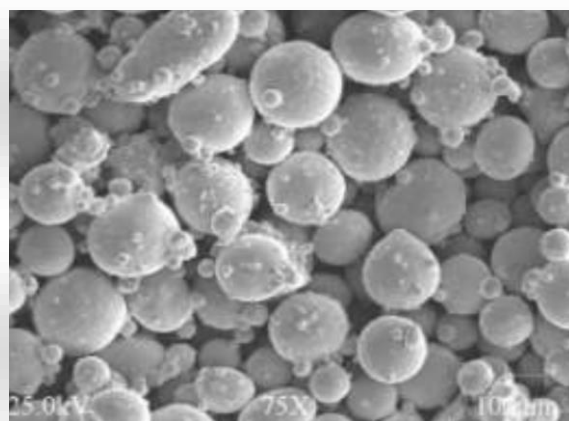
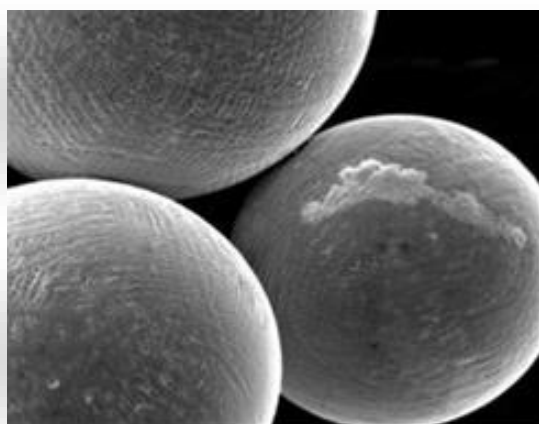
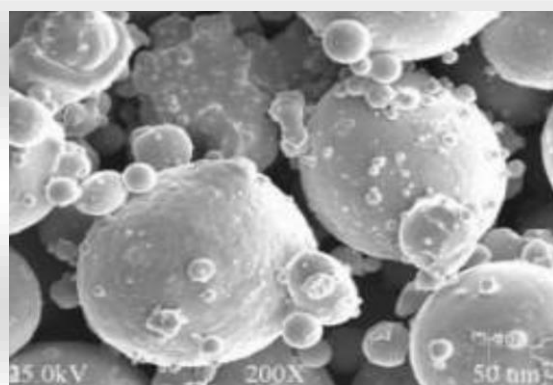
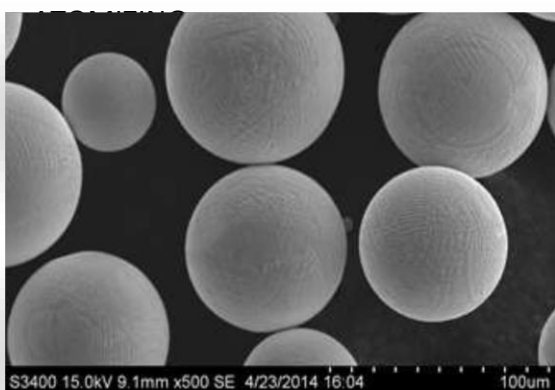
Advantages of the granules obtained by the PREP method:

- Productivity up to 1500kg per technological cycle;
- Dimension – F20-150mkm.
- Low content of gas impurities (O₂, N₂, H₂)
- Sphericity and narrow fractional composition of granules
- High shaking density (up to 68%)
- High quality granules
- Faster transition to other metals



Differences of granules obtained by PREP and Atomizing methods:

PREP



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